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RESEARCH-ARTICLE

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Exploring the Attacks, Impacts, and Mitigations in a Real-Time Streaming Protocol Service of IP Cameras

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ABSTRACT

The constant need for security has paved the way for the use of IP Cameras and CCTVs. Given that everyone now has an internet, IP cameras have become more popular as it allows Real-Time Streaming Protocol (RTSP) which grants the user video and audio streaming over the internet. This allows for pausing, playing, and recording video and audio data for later usage. Despite its popularity, RTSP has multiple vulnerabilities which pose a security risk for its users. Although multiple studies are exploring these risks, there is a lack of understanding about the specific ways that these vulnerabilities are exploited. This paper discussed the possible specific attacks, impacts, and mitigations that could occur on an RTSP-enabled IP Camera. Based on the gathered studies, four vulnerabilities were found: (1) Unencrypted URL of the Streaming Server; (2) Unencrypted User Account, Wi-Fi, and RTSP credentials; (3) Unsecured System and Lack of Security Measures for Input Validations for RTSP requests; and (4) Improper Quality Assurance of Firmwares. These vulnerabilities lead to attacks such as Packet Sniffing, Man in the Middle Attacks, Denial of Service, and Brute Force Dictionary Attacks. Additionally, the impacts of these attacks were explored as well, such as loss and theft of data, service disruption, and brand reputational risks. Mitigations for these attacks and exploited vulnerabilities include the following: Utilizing OpenSSL to provide encryption and authentication of data over the networks, Implementing Cryptography-based APIs in the system development, Implementing an Access Control List to control authentication, changing default passwords to strong passwords, updating softwares and firmwares to ensure patches are applied, and making use of secure authentication modules that implement hashing algorithms, QR codes, and other authentication values.

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CCS CONCEPTS

• **Security and privacy**; • **Human and societal aspects of security and privacy**; • **Privacy protections**;

KEYWORDS

RTSP, IP Camera, Vulnerabilities, Cyberattacks

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1 INTRODUCTION

1.1 Background of the Study

Security is one of the major concerns for most Filipinos. Going around the Philippines, one could prove this claim by seeing the abundance of security guards and cameras in almost every house, store, or building. In the 21st century, the need for constant security and surveillance has led to innovations in automated surveillance technologies such as CCTV Cameras and IP Cameras. IP cameras are widely used in commercial and residential settings for surveillance and security purposes. They have become more convenient than analog CCTV cameras due to the emergence of the internet and their ability to be connected to various network connections as opposed to analog CCTV cameras, which can only transmit audio and video through a Digital Video Recorder [1].

However, together with the abundance of IP cameras in various locations, so too are the concerns in regard to the vulnerability concerns of these devices. In 2017, there was a published list of every vulnerability involved with several IP camera makers such as Samsung, Hikvision, Cisco, Panasonic, and many more [2]. Due to the ease of access of IP cameras to the internet and various networks, IP cameras have become one of the primary targets of attackers for vulnerability exploits and attempt to gain unauthorized access to these devices; in which one of the vulnerabilities that this topic is going to focus on is from the Real-Time Streaming Protocol or the RTSP service.

Many IP cameras, media servers, and other devices employ RTSP, created by the IETF (Internet Engineering Task Force), to broadcast audio and video over the Internet. It enables clients to remotely control multimedia streaming, including music and video, by sending server commands like "play," "pause," and "record." [3].

The industry has widely embraced RTSP. However, despite the popularity and convenience of the RTSP, it can be easily used by attackers to obtain unauthorized access to IP cameras or to intercept and monitor the video and audio feeds, which makes it a security issue. Therefore, it is crucial to secure the RTSP service by understanding and providing strong mitigation and safeguarding methods to prevent unauthorized access and protect the data stream.

1.2 Research Questions

Despite the existing studies that have examined the security of RTSP, there is a lack of understanding of all the vulnerabilities present in an RTSP-service-enabled IP camera, the attacks that threat agents can use to exploit the vulnerability, the potential impacts of the vulnerability, and the most effective security measures to mitigate the risk, and this study provides an attempt to fill in that gap. The study aims to answer the following questions:

- What are the specific ways that attackers can exploit the RTSP service vulnerability in IP cameras, and how are they done?
- How can the potential impacts of the RTSP service vulnerability affect organizations and individuals?
- What are the most effective security measures and mitigation techniques that can be implemented to mitigate the risks associated with the RTSP service vulnerability in IP cameras?

1.3 Objectives of the study

The research questions aim to fulfill the following objectives of the study:

- To identify the specific ways how attackers can exploit the RTSP service vulnerability in IP cameras and analyze the process of exploiting it.
- To understand the potential impacts and damage of the RTSP service vulnerability on organizations and individuals.
- To provide a comprehensive literature review of existing studies and research on RTSP service vulnerability to provide practical recommendations to improve the security measures of the IP camera system.

1.4 Significance of the Study

This study allows organizations and individuals to understand the risks associated with the RTSP service in IP cameras and allows them to improve the security of their camera systems which can help them better protect their system and data from unauthorized attacks and breach.

1.5 Scope and Limitations

The study will focus on the RTSP service vulnerability in IP cameras, specifically on how attackers can exploit the vulnerability, the potential impacts of the vulnerability, and the security measures that can be implemented to mitigate the risks. The study will include a review of existing studies and research on RTSP service vulnerability. Lastly, the study will also provide practical recommendations for organizations and individuals looking to improve the security of their IP camera systems.

However, the study will not cover other types of vulnerabilities or attack vectors that may affect other types of devices. The study will only be based on the IP cameras that use the RTSP protocol and will not cover other protocols. The study will not cover the testing of the mitigation techniques and technologies; it will only cover the existing studies and research that have already been conducted. This study will not also be specifying and will not be focusing on an IP camera brand; however, it may be used as an example on how the attack was done. Ultimately, the study will be limited to the information available in the literature; it may not cover all the possible attacks, impacts, and mitigations

2 REVIEW OF RELATED LITERATURE

To fully analyze the attacks and impacts that the vulnerability from the Real-Time Streaming Protocol Service creates, it is essential to compile several resources related to the study. This section focuses on discussing related literature and studies to address the objective set in this study. Credible references from IEEE explore, ACM, Scopus indexed databases, etc. were gathered and are compiled in this paper to create an in-depth analysis of the topic.

2.1 Overview about IP Cameras and Real-Time Streaming Protocol

Internet Protocol Cameras, also known as IP cameras, belong to a completely digital video network system that are connected to the network and into the PC for viewing and storing digitized surveillance materials via a network switch, typically a CAT5e cable. They are usually described as a camera and computer merged into one device that automatically digitalizes the video material, as opposed to an Analog CCTV Camera that needs an external encoder to digitize videos [4]. Figure 1 shows a typical structure of an IP camera composing of a lens, filter, sensor, a chip processor that controls most of the function of the IP camera such as providing the number of pixels to provide a high-quality camera, and the ethernet interface to be connected to the network either via Ethernet, LAN, or WLAN.

IP cameras may also be monitored via an application through mobile or web, if they are connected within a network, therefore enabling the viewers to record, view, and save videos for future purposes [4]. IP Cameras also present multiple benefits and differences over analog cameras over the years due to emerging technologies and the change in demand for a clearer and a more scalable access to surveillance materials. Table 1 presents the comparison between the characteristics of an Analog Camera and an IP Camera as presented by Hahne [2022] [5], which is supported by the study of Popovic et al. [2013] [4].

2.2 Vulnerability Analysis

IP Cameras are no exemption to vulnerabilities. Since IP Cameras are connected to the internet, it is extremely more exposed to threats than CCTV Cameras due to the grave number of threats present on

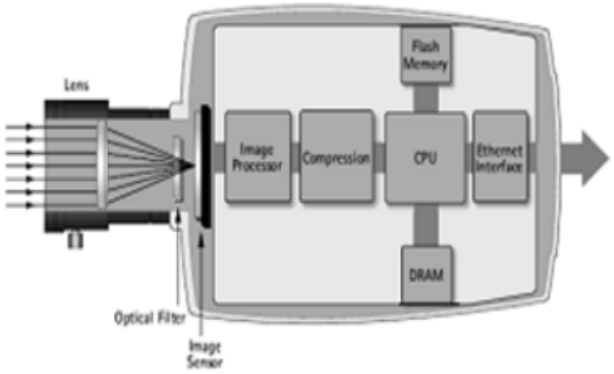

Figure 1: Structure of an IP Camera by Popovic et al. [4]

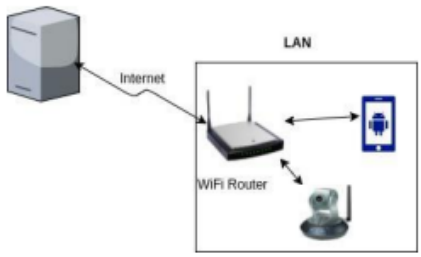

Figure 2: Connection of the IP Camera [6]

the internet with just a simple mismanagement of a server. Other than mismanagement of the server controls, other vulnerabilities include broken cryptography, lack of security protection such as binary protections, misconfigured hardware components, and bad implementation of authorization and authentication services [6]. On the other hand, Real-Time Service Protocol, despite its high usage of IP cameras to stream video and audio data, still presents several vulnerabilities, especially if it is not properly configured. These vulnerabilities can be exploited by threat attackers who want to gain unauthorized access to video and audio data to gather information about a specific area or event.

An example of a vulnerability found in an IP camera with an RTSP Service enabled is in the study by Seralathan et al. [2018]. The researchers used a generic IP camera to find a vulnerability in its security. Figure 2 shows the setup of the IP Camera in which the IP camera is connected to the LAN via a WiFi Router with a registered mobile device, to see what the IP Camera streams, which is also connected to the same router. The WiFi Router is then connected to the internet. All other communication will happen in the server of the IP camera once it is connected to the internet [6].

For the security context of the IP Camera, the researchers noted that adding any other device to the mobile server is difficult to get the video stream as it requires a sort of authentication. Afterward, the researchers used a Network Mapper scanning tool to scan for open ports and identify the services currently running in the devices and found two ports: (1) Port 554/TCP RTSP; and (2) Port 5000/TCP

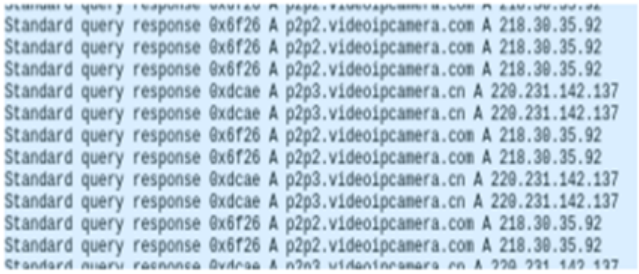

Figure 3: WireShark Data of IP Cameras [6]

SOAP. With that, the researchers were able to capture all network packet data using the Wireshark tool. They captured the data both from the mobile and from the camera. Figure 5 shows the setup they did to capture the data from the camera by using a network tap which duplicates the data flowing from both ends of a network, while Figure 6 shows the data captured from the IP Camera.

Based on the topology and data captured by WireShark shown in figure 5, the researchers observed that all the streaming servers of the IP camera follow the format "p*p*.videoipcamera.cn" or "p*p*.videoipcamera.com". This means that any number from 1 to 6 is acceptable by the server, and you will already gain access to the streaming server [6].

Another observation that the researchers noticed during their data capturing was that the Wi-Fi Credentials and the Account Credentials were stored in clear text. Basically, the RTSP URL and the credentials to the Wi-Fi and account are all stored in unencrypted texts and have no cryptographic hashes. Hence, the researchers were able to conclude that one vulnerability of RTSP-enabled cameras is Unencrypted Communication which can lead to attacks such as Brute Force Attacks and Man-in-the-Middle Attacks [6].

Another vulnerability that RTSP-enabled service in an IP camera is the ability to have a buffer overflow through Distributed Denial of Service Attacks or DDoS. A study by Prokofiev, Smirnova, & Silnov [2017] discussed how DDoS is used as an attack to gain access to video streams [7]. Attackers use port 554, the port that an RTSP service uses to listen and accept requests. Attackers use port 554 to send multiple requests, particularly to the OPTION and DESCRIBE requests of the RTSP service. A buffer overflow vulnerability is present in an RTSP-enabled service IP camera if the software/system being used is not secured enough and does not have enough input validations to filter out requests. If this is the case, cybercriminals can exploit this vulnerability to conduct a successful denial of service attack [7]. The buffer overflow vulnerability is also present in the IP camera in an attack on one of the products of Shenzhen Neo Electronics due to the improper quality assurance of the firmware level of their products. This vulnerability was then exploited via the DESCRIBE Request and the NVR Gateway web server [8].

Finally, the vulnerabilities coincide with those involved in an RTSP-enabled service IP camera taken from Manske's works [2019]. Table 1 describes a list of vulnerabilities discovered in an IP Camera [9]. For this study, the researchers decided to only include the RTSP-related vulnerabilities.

Table 1: Discovered Vulnerabilities in an IP Camera [9]

Number	Vulnerability	Result of the Vulnerability
1	Default user and RTSP Credentials	Weak authentication method
2	Unencrypted RTSP messages	Threat to data leakage
3	Unencrypted RTSP traffic	Threat to data leakage

Overall, the vulnerabilities present in an RTSP-enabled service in an IP camera based on the gathered studies are the following: (1) Unencrypted URL of the Streaming Server; (2) Unencrypted User Account, Wi-Fi, and RTSP credentials; (3) Unsecured System and Lack of Security Measures for Input Validations for RTSP requests; and (4) Improper Firmware's. Quality Assurance. With these vulnerabilities, several attacks can be made possible by threat attackers to gain access to the streaming services of the IP Camera. The next part of the study will discuss possible attacks with the given vulnerabilities.

2.3 Man-In-The-Middle – Packet Sniffing

In a study by Seralathan et al. [2018], a MITM attack can be made by having unencrypted communications between the IP Camera and the mobile phone. Unencrypted communication is the transmission of data or information over a network without encryption. The unencrypted communications expose the sensitive information being sent from the mobile phone. Additionally, unencrypted communications mean all data transferred between the mobile phone and the IP camera are all readable in plain and clear text. Transferring data without encryption threatens to undermine not just the security of the camera but also that of the LAN on which it is installed [6]. In the case of the study, the mobile application from the phone inputs the Wi-Fi SSD and other credentials to connect the IP camera to the Wi-Fi. When the unsuspecting user inputs the credentials without encryption, an attacker can place the device between the mobile phone and the IP camera to perform a MITM attack. An attacker can use the Wireshark application to capture the credentials in clear text. The Wireshark filter "Follow TCP stream," which chooses all the TCP packets of the specific packet chosen and dumps all of their contents, was used to retrieve the data that contains the Wi-Fi identifier and password [6].

2.4 Mitigation and Recommendation Analysis

RTSP and IP cameras are vulnerable to various types of cyberattacks that can compromise the security of the cameras, steal sensitive information, or disrupt the regular operation of the cameras. However, there are ways to prevent and mitigate attacks to reduce the damage and potential loss of valuable data. Mitigating these attacks is crucial to protect the cameras and their information.

For MITM attacks, the study of Seralathan et al. [2018] suggests utilizing well-known OpenSSL libraries [6]. OpenSSL is an open-source implementation of the Secure Sockets Layer (SSL) and Transport Layer Security (TLS) protocols. These protocols provide secure communication over networks, such as the Internet, by encrypting data and authenticating the identity of the parties involved [10]. An OpenSSL encrypts all communications between the IP camera and mobile application [6]. Furthermore, Vennam et al. [2021] suggests the utilization and enhancements of API, particularly cryptography-based API. [11] This will ensure that secure functionalities will be utilized by developers in developing applications for IP cameras examples include Secure Socket Layer (SSL) and Transport Layer Security (TLS) protocols.

DoS attacks can be mitigated through the implementation of access control and proper authentication of users. Access control specifies the limits and capabilities that specific users or lists of users can utilize, in this case, resources and actions can be minimized among botnets controlled for a DoS attack [11].

Another mitigation technique is to change default passwords to stronger passwords [11]. Changing the default passwords reduces the chances of security breaches from Brute Force attacks [12]. Another study by Yildrim and Machie [2019] on creating strong passwords should include at least eight characters long and contain at least one uppercase, one lowercase, one number, and one special character. In addition to strong passwords, they should be memorable and not be used multiple times in other accounts [13]. It is also recommended to constantly change passwords every 90 days from the day it was changed. In general, it has always been a safe and simple practice among IT users to always keep their systems updated to the latest versions. This eliminates the possibility of any kind of vulnerabilities, defects, errors, among systems to remain unpatched or unfixed throughout the duration of usage [11]. This eliminates the possibility of any kind of vulnerabilities, defects, errors, among systems to remain unpatched or unfixed throughout the duration of usage

A proposed authentication system was published towards mitigating the possibility of brute force attacks in surveillance monitoring systems. An enhanced authentication system will distinguish legitimate requests from illegitimate requests that cause flooding [11]. The authentication system has additional steps in registration and logging in that provides more authentication than traditional username and password only. During registration, an additional step from traditional username and password is uploading/selecting an image for authentication, a combination of hashing algorithms MD5 and SHA1 will generate a hash based on the image uploaded which will be stored in the database. During logging in, the user will have to select his/her uploaded image and input the username and password. If the inputs are correct, a QR code will be generated which will be scanned through a mobile app, this QR code will provide a value that will be required during the login page, entering the correct value will yield a successful log in [14].

3 CONCLUSION

The vulnerabilities present in an RTSP-enabled service in an IP camera based on the gathered studies are the following: (a) Unencrypted URL of the Streaming Server; (b) Unencrypted User Account, Wi-Fi, and RTSP credentials; (c) Unsecured System and Lack of Security Measures for Input Validations for RTSP requests; and (d) Improper Quality Assurance of firmware. Through these vulnerabilities, threat attackers were able to find attacks that exploits these said vulnerabilities.

There are three ways where an attacker can exploit RTSP service vulnerabilities in IP cameras: First is Packet Sniffing. Packet sniffing can be done using a network analysis application like Wireshark. Wireshark can capture packets wherein an attacker can retrieve a packet and dump the contents of the packet. It can also be used for RTSP services without any encryption. If the RTSP does not have any encryption, the dumped packet may contain the necessary credentials in plain text to access the victim's Wi-Fi and camera. Second is the Denial-of-Service Attack. This can be performed by placing the attacker's device between the connection of the target and the camera. The attacker can use software that can intercept and manipulate network traffic, such as Ettercap. The attacker can immediately gain access and shut down all streaming services by killing the connection connected to the RTSP port, making the service unavailable to the consumers which could potentially be more damaging. Lastly, a brute force attack can be performed on RTSP-service enabled IP cameras that have default security configurations. An attacker can use a brute-force dictionary that contains a list of words that are commonly used as usernames and passwords. Additionally, an attacker can use a brute force application tool such as Hydra can launch an automated attack against a target system or application, attempting to guess the correct password or other credentials.

Individuals who use streaming services that support RTSP are at risk of being spied upon by threat agents via IP cameras. Attackers can gather personal and sensitive information via monitoring surveillance such as daily behaviors, activities, etc. that can be used for personal gain or identity theft. In an organizational perspective, this would mean the exposure of admin authentication details, security routines, and financial details that are otherwise confidential. For individuals, their personal data such as daily routines, biography data, and authentication details could be leaked to the attacker. Furthermore, unauthorized access to IP cameras via RTSP vulnerabilities serves as a risk to one's privacy and safety.

On the other hand, organizations attacked by a DoS are at the risk of losing revenue since the cyberattack halts and significantly slows down the services that the organization provides. Additionally, resources are wasted on solving the issue, which could have otherwise been prevented through better security implementations. Also, having vulnerabilities towards RTSP puts brand reliability and reputation at risk. Once information is dispersed of how manufacturers' products contain potential vulnerabilities, this will significantly impact consumer's buying choices. In essence, a conglomerate that has been successfully infiltrated by a cyber attacker puts itself in a state where future threats could see the successful attack as an encouragement or steppingstone for future attacks. This factor can impact overall profit making and growth of an organization.

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